Determining Profiles of Interest in Educational Offerings through Clustering Techniques

Final Project - Expert Program in Data Mining

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Agenda

1 Introduction and Objectives

2 Methodology and Project Plan

3 Exploratory Data Analysis

4 Clustering Models

5 Predictive Models

6 Implementation and Recommendations

7 Conclusions and Recommendations

Project Context

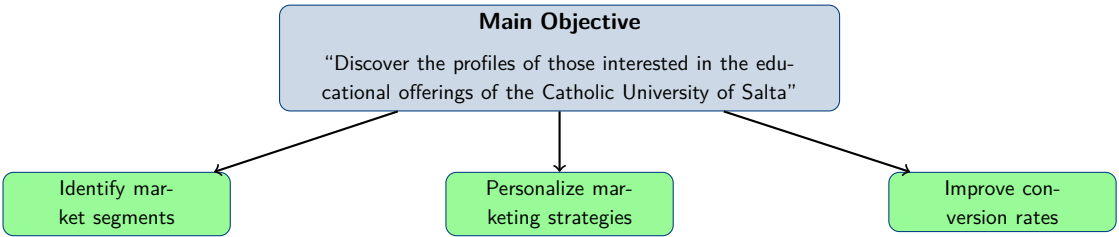
Problem

- The Marketing department needs to better understand profiles
- Current data only contains “cold” variables
- Need for segmentation with “soft” variables

Proposed Solution

- Apply unsupervised clustering techniques
- Avoid subjective segmentations
- Discover hidden patterns in data

Business Objectives



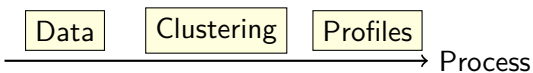
Goal

Achieve at least **two low-fuzziness clusters** with clear interpretation

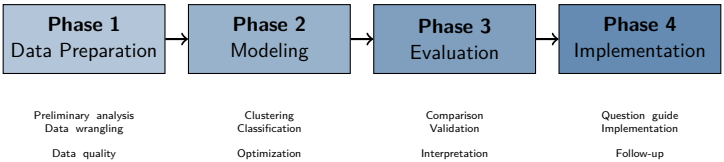
Data Mining Objectives

1. **Descriptive Objective**
   - Achieve potential client profiles through clustering
   - Identify distinctive characteristics of each segment

2. **Predictive Objective**
   - Predict the profile of new interested parties
   - Improve personalized academic offerings



Project Plan - Phases



Selected Modeling Techniques

Stage	Techniques
Stage 1: Exploratory Analysis	Correlation Matrix Principal Component Analysis (PCA)
Stage 2: Clustering	Hierarchical Clustering K-means Latent Class Analysis (LCA) DBSCAN (Density-based Clustering)
Stage 3: Classification	Decision Trees (CART) Random Forest

Selection Criteria

Techniques were chosen that allow both **interpretability** of results and **accuracy** in

Data Description

Dataset Characteristics

- **1,822** total records
- **17** variables per record
- Data from surveys
- Multiple responses allowed

Main Variables

- **Demographic**: Age, Gender
- **Preferences**: Study modality
- **Communication**: Social networks
- **Desired information**: Student life
- **Academic**: Faculties of interest

Data Description- Continuation

Challenges Encountered

- UTF-8 encoding without BOM
- Multiple response separator
- Missing values and need for transformation

Correlation Analysis

Positive Correlations

Variables	r
TEST_STUDENTS - TEST_GRADUATES	0.51
STUDENT_LIFE - EXCHANGE	0.42
AGE_18_25 - IN_PERSON_MODE	0.40
AUDIO_FORMAT - VIDEO_FORMAT	0.34

Insight 1

Young people (18-25) show interest in student life and exchange

Negative Correlations

Variables	r
DISTANCE_MODE - IN_PERSON_MODE	-0.57
AGE_18_25 - DISTANCE_MODE	-0.33
SCHEDULE_INFO - DISTANCE_MODE	-0.32

Insight 2

Young people do not prefer distance learning

Principal Component Analysis

PCA Results

- The first 2 components explain only **24%** of the variance
- No variable clearly dominates the components
- Relatively uniform and low contributions

Component 1

- Student life information
- Exchange information
- Orientation: **Comprehensive academic interest**

Component 2

- Distance mode (+)
- Age 18-25 (-)
- In-person mode (-)
- Orientation: **Modality preference**

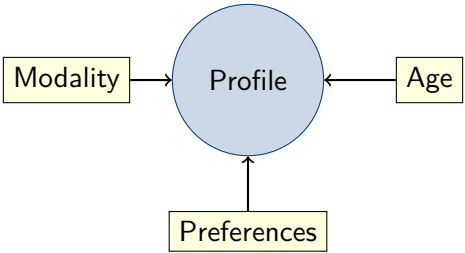
PCA Conclusion

PCA Conclusion

Three key factors profile the respondent:

- **Study modality**
- **Age**
- **Academic information preferences**

PCA Conclusion- Continuation



Clustering Model Comparison

Model	No. Clusters	Criterion	Fuzziness Index
Hierarchical	2	Intra-class inertia	**0.617**
K-means	2	Intra-class inertia	0.625
LCA	3	Minimum BIC	0.619
DBSCAN	Variable	Density	Not applicable

Decision

The **Hierarchical** model is chosen for presenting lower fuzziness

Non-Fuzziness Index

Non-Fuzziness Index

$$\text{NFI} = \frac{\sum_{k=1}^{m} \sum_{i=1}^{n_k} |rf_{1k}(C_i) - rf_{2k}(C_i)|}{m \cdot (s-1)} \tag{1}$$

where:

- m = number of questions
- n_k = categories of question k
- s = number of clusters
- $rf_{jk}(C_i)$ = relative frequency

Non-Fuzziness Index - continuation

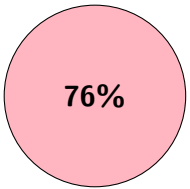
Interpretation

- Values close to 0: very fuzzy clusters
- Values close to 1: well-differentiated clusters
- Obtained value 0.617: acceptable differentiation

Identified Profiles - Profile 1

Profile 1: Adults (76%)

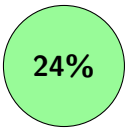
- **Age**: Over 25 years
- **Contact**: WhatsApp and email
- **Networks**: Facebook
- **Modality**: Distance
- **Faculties**: Law, Economics, Education
- **Not interested in**: University life



Identified Profiles - Profile 2

Profile 2: Young People (24%)

- **Age**: 18-25 years
- **Contact**: WhatsApp and email
- **Networks**: Instagram and Facebook
- **Modality**: In-person/Distance
- **Faculties**: Law, Economics
- **Interested in**: University life



Validation

Clusters are interpretable with more than two variables with clear interpretation

Decision Trees - Initial Challenge

Initial Challenge

- CART algorithm with 70% training - 30% testing
- Problem: Different trees in each execution
- Temporary solution: Reduce testing to 10%
- Persistent problem: Instability

Trade-off

We sacrifice precision for interpretability and stability

Decision Trees - Adopted Strategy

Adopted Strategy

1. **Tree O (Original)**: Pruning with restrictions
   - Maximum depth: 5 levels
   - Minimum for split: 100 cases
   - Objective: Interpretability (insight tree)
2. **Tree A**: Cross-validation with caret
3. **Tree B**: Only less fuzzy variables

Random Forest - Important Variables

Variable	Importance
INFO_EXCHANGE	22.63
FORMAT_PHOTOS	20.71
AGE_18_25	18.40
WANT_VISIT	17.66
INFO_STUDENT_LIFE	16.57
NETWORK_INSTAGRAM	15.91
IN_PERSON_MODE	15.67
INFO_SCHEDULES	15.39
DISTANCE_MODE	14.99
INFO_STUDENT_TESTS	14.72

Predictive Model Comparison

Metric	Tree A	Tree B	Tree O
Variables used	7	4	4
In top 10 RF	7	2	**3**
Interpretability	Low	High	**High**
Precision	82%	78%	**79%**

Final Decision

Tree O is chosen for balance between precision and interpretability

Decision Rules for Classification

Rule	Profile	Recommended Offers
Age 18-25; Interested in exchange; Wants multimedia info	**Young**	In-person and distance programs
Age 18-25; NOT interested in exchange	**Adult**	Distance programs + Teacher training

Decision Rules for Classification (cont.)

Rule	Profile	Recommended Offers
Age 18-25; Interested in exchange; NO multimedia; Wants schedule info	**Young**	In-person and distance programs
Age over 25; Does NOT want schedule info	**Adult**	Distance programs + Teacher training

Practical Implementation

These rules can be implemented in a CRM system for automatic classification

Factor Analysis - Additional Validation

Identified Factors

1. **Factor 1**: Young people with comprehensive interest
   - Student life, exchange, activities
   - Maximum factor loading: 0.537
2. **Factor 2**: Adults distance modality
   - Distance modality, not in-person
   - Maximum factor loading: 0.568
3. **Additional factors**: Gender and contact preferences

Result Convergence

Factor analysis **confirms** the profiles identified by clustering

Why Interpretable Models?

Legal and Ethical Mandates

- Transparency in automated decisions
- "Right to explanation" (GDPR)
- Avoid algorithmic bias
- Corporate responsibility

Business Benefits

- Stakeholder trust
- Implementation ease
- Error detection
- Continuous improvement



Black Box

✗



Interpretable

✓

Final Conclusions

Project Achievements

1. ✓ Successful identification of **2 clearly differentiated profiles**
2. ✓ Reduction of fuzziness to acceptable levels (0.617)
3. ✓ Interpretable predictive model with reasonable precision (79%)
4. ✓ Cross-validation between different techniques

Marketing Recommendations

Adult Profile (76%)

- Focus campaigns on distance learning
- Use Facebook as main channel
- Highlight schedule flexibility

Young Profile (24%)

- Promote university life and exchanges
- Use Instagram in addition to Facebook
- Offer multimedia information

Future Work

Technical Improvements

- Incorporate more soft variables
- Apply deep learning techniques
- Temporal validation of profiles
- Survival analysis

Business Applications

- Automatic scoring system
- Landing page personalization
- Advertising budget optimization
- Dropout prediction

Expected Impact

Expected Impact

15-20% increase in conversion rates
through profile-based personalization

"The ability to explain or present in human-understandable terms the predictions of an Automated System"

Thank you!

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**“Data has stories to tell,
we just need the right tools
to listen to them”**